Measurement: Sensors 39 (2025) 101876

Contents lists available at ScienceDirect

Measurement: Sensors

journal homepage: www.sciencedirect.com/journal/measurement-sensors

Enhancing acute leukemia classification through hybrid fuzzy C means and random forest methods

K. Lakshmi Narayanan [a], R. Santhana Krishnan [b], Y. Harold Robinson [c], S. Vimal [d,*], Tarik A. Rashid [e], Chetna Kaushal [f], Md. Mehedi Hassan [g,**]

[a] Department of Electronics and Communication Engineering, Francis Xavier Engineering College, Tirunelveli, India
[b] Department of Electronics and Communication Engineering, SCAD College of Engineering and Technology, India
[c] Department of Computer Science and Engineering, Francis Xavier Engineering College, Tirunelveli, India
[d] Department of Artificial Intelligence and Data Science, Sri Eshwar College of Engineering, Coimbatore, India
[e] Department of Computer Science and Engineering, University of Kurdistan Hewler, Erbil, Iraq
[f] Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India
[g] Computer Science and Engineering Discipline, Khulna University, Khulna, 9208, Bangladesh

ARTICLE INFO

Keywords:
Acute lymphoblastic leukemia
Lymphocytes
Machine learning
Image analysis
Random forest classifier

ABSTRACT

Leukemia is a category of cancer that is normally found in blood and bone marrow, and which causes rapid abnormal development in the making of white blood cells than the required amount. The produced white blood cells could be ineffective to fight against harmful infections and can even prejudice or restrict the capability of the bone marrow to generate red blood cells and blood platelets. If this is not diagnosed in the earlier stage, it may start to affect the function of the internal organs and cause death. Normally, entire blood counts image analysis and diagnosis are done manually which is an inaccurate and time-intensive process. In this proposed method the classification is tested with two Machine Learning algorithms which are Hybrid Fuzzy C Means (FCM) and Random Forest algorithm (RF) and Support Vector Machine for the detection and classification of Acute Leukemia disease and their performance was evaluated. The dataset comprised of 8637 images which included infected images, normal images and augmented images from different dataset providers and RGB to CMYK conversion with histogram equalization is applied for pre-processing, K means for Image Segmentation. Experimental results convey that Hybrid FCM and RF Algorithm attained an accuracy of 99.06 %, a sensitivity of 99.4 %, and a specificity of 97.8 % respectively, and the ROC (Receiver Operating Characteristic) curve shows that the result produced by the Hybrid FCM & RF based Classifier is best suitable in diagnosing the classification of the Acute Leukemia disease. The tool used for developing the proposed method was Matlab R2018 software.

1. Introduction

Acute lymphoblastic leukemia (ALL) is a type of cancer that affects the blood and bone marrow, resulting from the rapid proliferation of immature lymphocytes, which are a form of white blood cell. It is categorized as a type of acute leukemia, meaning it progresses quickly and requires immediate diagnosis and treatment. It reduces the creation of usual red blood cells and platelets consequently it leads to anemia [1]. The uncharacteristic leukocytes will spread to the normal human blood and will capture the different parts similar to the liver, grumpiness, kidney, and brain. Leukemia can either be mainly classified as acute or chronic Leukemia. The classification depends upon the progressing speed. Each of these two types of Leukemia has two subclasses [1] which are demonstrated in Fig. 1.

Further, this Leukemia could be categorized into two types according to the size and shape of the White blood cells they are Lymphoblastic or Myelogenous. If the unhygienic cells are granulocytes and monocytes then they are classified as Myelogenous, if the affected cells are

This article is part of a special issue entitled: Wearable Innovations for Human Augmentation published in Measurement: Sensors.

* Corresponding author.
** Corresponding author.
E-mail addresses: klnarayanan@francisxavier.ac.in (K. Lakshmi Narayanan), santhanakrishnan86@gmail.com (R. Santhana Krishnan), yhrobinphd@gmail.com (Y. Harold Robinson), svimalphd@gmail.com (S. Vimal), tarik.ahmed@ukh.edu.krd (T.A. Rashid), chetnakaushal3558@gmail.com (C. Kaushal), mehedihassan@ieee.org (Md. Mehedi Hassan).

https://doi.org/10.1016/j.measen.2025.101876
Received 2 April 2024; Received in revised form 4 April 2025; Accepted 5 April 2025
Available online 7 April 2025
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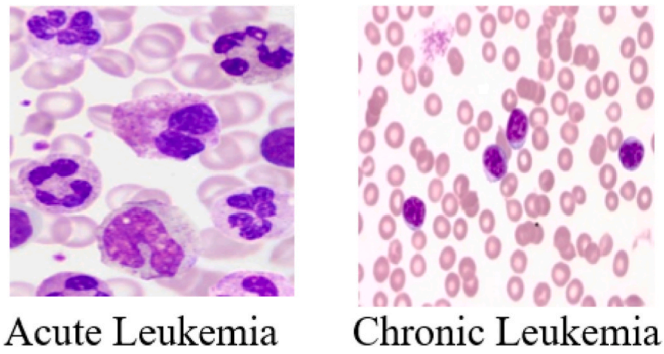


Fig. 1. Acute and chronic leukemia.

lymphocytes, then it is categorized as lymphoblastic [2]. Acute Lymphocytic Leukemia (ALL)is a kind of blood cancer that will form in the bone marrow which is the inner part of the bone [3]. This ALL will be formed on the lymphocytes a kind of RBC that takes care of the immune system There are three main types of lymphocytes, B Lymphocyte which is responsible for producing antibodies, T Lymphocytes which are involved in cell-mediated immunity and Natural Killerwhich play a key role in the innate immune response by targeting and destroying virus-infected cells and tumor cells. Acute Myeloid Leukemia (AML) is the next kind of blood cancer that will also form in the bone marrow but also moves faster into the blood cells. It occasionally spread into other body parts such as lymph nodes and the central nervous system. This AML is also called acute granulocytic leukemia [4]. Chronic Lymphocytic Leukemia (CLL) is another type that progresses very slowly when compared with any other type of Leukemia. This is the most common type of Leukemia which affects older adults. People with this Chronic Lymphocytic Leukemia will experience very frequent infections. This type of Leukemia will increase the risk of getting other forms of cancer [5]. Chronic Myeloid Leukemia (CML) will be found in blood-forming cells of bone marrow. This is also a common type of Leukemia that affects elder adults. This type of leukemia produces many white blood cells and slowly starts crowding in the bone marrow and interfering with the usual WBC [6]. All the subtypes of Leukemia are demonstrated in Fig. 2.

Different types of medical image classifications using machine learning introduce a new research field of early detection of destructive diseases which reduces the manual system incorporation. The main objectives of this paper are.

1) The major objective of the proposed system is to apply a hybrid machine learning technique to distinguish Acute Lymphocytic Leukemia detection from the microscopic blood image.
2) To incorporate feature extraction in SVM classification, VCC16 Convolutional Neural Networks (CNN) with ImageNet dataset is used [7].
3) To incorporate feature extraction in Random Forest classification, A Fuzzy C Means Clustering is incorporated.
4) The efficiency of both classification methods is compared based on classification parameters.
5) The proposed system is compared with other state-of-the-art machine learning algorithms.

2. Related works

A hybrid approach [8] has been implemented for classifying WBC Leukemia as the computational training and development cost of the Convolutional Neural Network is high. Here the features are extracted initially from WBC VGGNet which is an influential CNN Framework and pre-trained on ImageNet the features extracted are filtered through the Enhanced Slap Swarm Algorithm which removes the noise signals. This method was tested with two publicly available datasets which show enhanced accuracy and minimized computational complexity. An image processing technique [9] has been used to detect the white blood cells in Acute Myeloid Leukemia, Acute Lymphoblastic Leukemia, and Myeloma here they have used YCbCr for making the image suitable for segmentation, and for segmentation, they have used K means clustering algorithm the feature extraction is done through GLCM and the final classification of the disease was done using the Convolutional Neural network the overall accuracy obtained by their method is 97.3 %. The steps are involved in the classification of leukemia by comparing various image processing techniques and giving the pros and cons of various image processing techniques [10]. A Convolutional Neural Network based method [11] has been utilized for the identification of Leukemia and its subtypes. This method achieves an accuracy of 88 % in disease detection and 81 % in classifying its subtypes and shows its superior performance than other algorithms such as Naïve Bayes, KNN, and SVM.

A similar CNN-based classifier [12] for detection of the Leukemia archives an accuracy of 99 % both with ROC and AUC. The enhanced technique [13] has been compared with several machine learning algorithms for identifying Acute Lymphoblastic Leukemia such as SVM, Logistic regression, decision tree, and random forest classifiers experimental results show all the algorithms show results higher than 90 % of which SVM shows its superiority towards the detection of leukemia. But it is not possible to analyze the efficiency of the classifier based on the classification algorithm alone but it needs to have to look into how many datasets they have used how the features are extracted, what type of segmentation algorithm was been used etc. only after considering all these factors can estimate the efficiency of a method. Image segmentation and data mining algorithms-based detection of Acute Lymphoblastic Leukemia [14] has been involved the results classify the ALL into three different classifications they are L1, L2, and L3 which are normal peripheral and abnormal blood smears with an overall accuracy of 98 %. A deep learning-enabled method [15] for the classification of Acute Lymphoblastic Leukemia has been utilized, the pre-trained Convolutional Neural Networks as a classifier for classifying leukemia and its subtypes. This method offers an accuracy of 99.00 % for disease detection and offers 96.06 % for classifying its subtypes. They also compared the results with various other deep learning algorithms such as SVM, KNN, and Hybrid Hierarchical Classifiers the results show the superior performance of CNN.

A Convolutional Neural Network based image classification method [16] has been used to distinguish the test image as a regular or abnormal blood image. Through this method, they have achieved an accuracy of

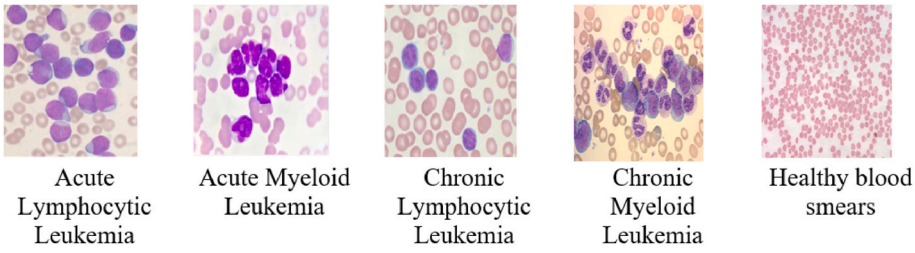


Fig. 2. Sub-types of leukemia.

96.6 % with an 1188 blood image dataset. A hybrid random forest-based image classification [17] is involved and suggests that the accuracy of the image classification should be enhanced effectively by selecting the optimal coefficients of the combined algorithm. And the proposed method was more stable and it is suitable for massive images.An improved method [18] has been used for detecting Acute Leukemia using a k-means clustering algorithm. This was implemented through basic image analysis, enhancement, and morphology-based techniques and achieved an accuracy of 92.8 %. They also tested the same with other algorithms such as a K nearest Neighborhood algorithm and Naïve Bayes Classifier which shows a very low accuracy. Several related techniques have been surveyed for Leukemia detection such as Support Vector Machines, K Nearest Neighborhood, sequential Neural Networks, etc. Most of the methods show accuracy above 90 % of which SVM shows quite higher accuracy than the other algorithms [19]. The image processing method [20] has been implemented for the identification of leukemia in human blood samples. Here the lab color spaced-based segmentation is used on the image to eliminate the unwanted portions in the image and extract the nucleus from the image. The extracted nucleus helps to take the shape-based feature and finally, KNN is used for classifier results showing the accuracy of this method is more than 90 %.

Medical image classification involves the use of deep learning techniques, particularly convolutional neural networks (CNNs), to analyze and categorize medical images [21]. This approach has significantly enhanced diagnostic accuracy and efficiency in various tasks, such as skin cancer detection and chest radiograph diagnosis. Despite its advancements, challenges like data scarcity, ethical concerns, and the need for interpretable models remain. The integration of artificial intelligence into clinical workflows is crucial for improving patient care and fostering trust in automated diagnostic systems. Medical image classification involves using deep learning models to analyze medical images for diagnostic purposes. The paper highlights that traditional transfer learning methods, often based on ImageNet, may not be optimal for medical tasks due to differences in image characteristics. It proposes a novel transferability metric that combines feature quality and gradient information, demonstrating that medical-specific datasets can outperform ImageNet in various classification tasks [22]. This research aims to enhance transferability estimation methods for improved medical image classification outcomes.

Medical image classification is a critical process in diagnostic decision-making that involves categorizing images to aid in disease detection and treatment planning [23]. The research paper presents a hybrid approach that integrates advanced image processing techniques, such as wavelet and Gaussian filters, with deep learning methods, specifically a CNN-LSTM architecture. This framework enhances image quality, optimizes feature selection using the Lion Optimization Algorithm, and ultimately improves classification accuracy, contributing significantly to advancements in medical image analysis and patient care.

A CNN-based Classification of Leukemia [18] has been used with features extracted by pre–a trained deep neural network which is a fine-tuned Alex Net with a dataset of 2820 images. A CNN-based enhanced system [24] has been used to detect Leukemia from the microscopic image analysis with seven different transformation techniques as data augmentation. Experimental results show the existing system has a very low accuracy of 88.25 and as the system is applied to the medical image for disease detection and classification the accuracy should be very high. From the literature survey, it is inferred that most of the research on the WBC blood sample classification was done majorly based on CNN, SVM, and KNN classification methods like MBAML [25], and HALM [17]. Research on the Random Forest and Fuzzy c means is relatively very less. This proposed system utilizes datasets and augmented datasets and altogether 8637 datasets and tested them with a Random Forest Classifier.

3. Materials and methods

The various details on the dataset collection and various steps involved in processing the image and finally applying the Random Forest algorithm to classify Acute Lymphocytic Leukemia are given.

3.1. Dataset

In this proposed work the datasets are collected from three different sources they are ALL IDB, Cancer Imaging Archive, and locally available diagnostic centers which are freely available sources and which contain microscopic images of both Acute Lymphocytic Leukemia and Healthy Blood smear (See Table 1). The datasets provided from these sources are for research purposes in the field of identifying and classifying Acute Lymphocytic Leukemia images. In these sources, Cancer Imaging Archive provides images for all types of Leukemia and ALL IDB provides a dataset only for Acute Lymphocytic Leukemia which is also considered reliable as per the suggestions given by the oncologist about the locations of the lymphoblast. Few images collected locally are mostly healthy microscopic images. ImageNet dataset is used for the experiments, and it is publicly available in https://paperswithcode.com/dataset/imagenet [7] which are 96x96 pixel in size of which 80 % of the images is used as training image and 20 % for testing.

3.2. Image augmentation

For any type of Machine learning research, the most important part is the data collection. In this case the data is the image as the research is on a medical application the datasets will not be available and most of them will be confidential. On the other hand, less data for training will lead to overfitting problems which will impact the performance of the system. Data Augmentation is one way through which is used to increase the dataset for any Machine learning algorithms. A variety of image alteration methods such as mirroring, shifting, and rotating, were adopted to acquire the different versions of the real dataset images. The proposed system utilizes the function augmented Image Datastore to increase the training image data by resizing, mirroring, and rotating.

Image resizing is a transformation that changes the size of an image by applying a scaling factor >1 (Increased by 150 %) or <1(Reduced by 50 %), Data Mirroring which is obtained by rotating the image from 0 to 360° to get additional training data. Image mirroring is done by both horizontal mirroring and vertical mirroring through which the image appear in left will be flipped to right and image in the top will be flipped to bottom [26]. Technically the original image data is transformed but for the system then it will be different data and that avoids the overfitting problem.

3.3. Image pre-processing

To progress the eminence of the input image and to facilitate the next processing steps pre-processing is required. The microscopic blood images may contain various types of noise in them and at the same time there may be some blurred regions in the image that is significant for the

Table 1
Dataset collection report from various sources.

Data Set	Type	No. of Infected	No. of Healthy	Augmented Image	Total
ALL IDB	ALL	178	175	3532	3885
Cancer Imaging Archive	ALL	90	–	1985	2075
Local Diagnostic Centre	General	65	50	2562	2677
Total					8637

detection. In the initial phase of image pre-processing the image is initially resized using a Matlab function imresize:

$$B = imresize(A, scale) \tag{1}$$

Where image B will be scaled times that of A, in this case, the image can be either RGB or binary image, later the image is filtered using the function imfilter.

$$B = imfilter(A, H) \tag{2}$$

This function filters the multi-dimensional array A with multi-dimensional filter H. Once the initial resizing and filtering are done the next phase is to convert the SRGB to CMYK and Histogram equalization. Convert the Standard RGB image to CMYK image to the best part of the WBC *n* the blood image.

$$C = makecform\ ('srgb2cmyk') \tag{3}$$

Histogram equalization is a most familiar technique to improve the dissimilarity of the image through which the contrast is improved and different lighting effects during the image acquisition will be overcome through this histogram equalization [27]. The image pre-processing steps are illustrated in Fig. 3.

3.4. Image segmentation

The next step of the classification process is image segmentation because segmentation is the progression of segregating the image into several parts called segments which helps in simplifying the representation of the image easy to understand and also provides meaningful information about the image to be classified. Some of the popular image segmentation techniques are edge detection-related functionality, thresholding technique cluster, and region-related methodologies. One of the very popular image segmentation techniques for medical images is the clustering technique. Clustering can be done in two different ways as Soft and Hard Clustering.

Hard clustering is a process of grouping the pixels in such a way that each pixel will be assigned to any one of the clusters. So that the member function fits into either zero or one to find whether the pixel fits into the cluster or not. Soft clustering is a development of grouping in such a way that data items can be available in multiple clusters at the same time when there is no need for binary memberships. This technique is more efficient than the other technique. The following are the different cluster-based image segmentation techniques available.

K means clustering belongs to the hard clustering category [28]. This is a type of clustering technique where the image is divided into k number of disjoint clusters. This clustering method has two phases. The initial phase is to find the k centroids. In the second phase, it obtains every point to the cluster that has the adjacent centroids from the corresponding data points. The most common method of finding the distance of the adjacent centroid is by Euclidean distance technique. Once the grouping was completed it re-computes the latest centroid of individual clusters based on the centroids and allocates the new points with the shortest distance to the cluster [29]. Thus, this algorithm is an iterative algorithm that reduces the overall distance from each object from the centroid, throughout the cluster in Fig. 4.

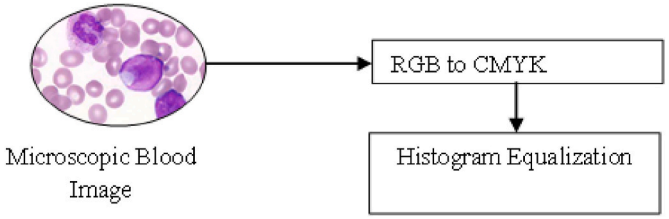


Fig. 3. Image Pre-processing steps.

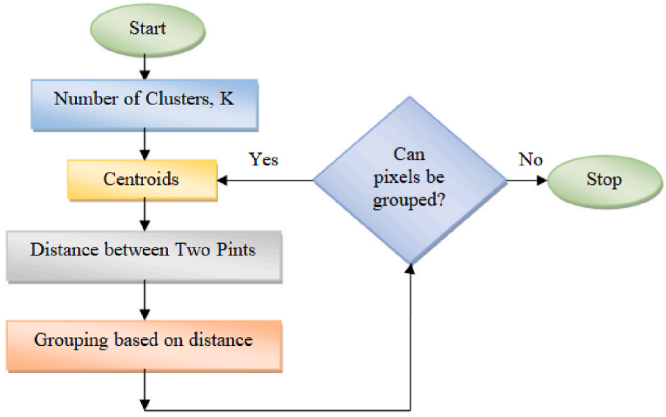


Fig. 4. K-means clustering.

Fuzzy c means a process of clustering that allocates a single pixel of the image to belong to several clusters which come under the category of soft clustering. This clustering type is very common in pattern reorganization and soft medical image segmentation, the reduction of the subsequent object function.

$$J_m = \sum_{i=1}^{N}\sum_{j=1}^{C} u_{ij}^{m}\left\|x_i - c_j\right\|^2 \tag{4}$$

Here m is the real number, u_{ij} is the membership degree, x_i is the calculated data, c_j demonstrates the d-dimensional cluster.

Algorithm: FCM (Fuzzy C Means)

Step 1: Set U=[u_{ij}] matrix, $U^{(0)}$
Step 2:Compute the centers vectors $C^{(k)} = [c_j]$ with $U^{(k)}$

$$c_j = \frac{\sum_{i=1}^{N} u_{ij}^{m} \cdot x_i}{\sum_{i=1}^{N} u_{ij}^{m}} \tag{5}$$

Step 3: Modify $U^{(k)}$, $U^{(k+1)}$

$$u_{ij} = \frac{1}{\sum_{k=1}^{c}\left(\frac{\|x_i - c_j\|}{\|x_i - c_k\|}\right)^{\frac{2}{m-1}}} \tag{6}$$

Step 4: If $max_{ij}\left\{\left\|u_{ij}^{(k+1)} - u_{ij}^{(k)}\right\|\right\} < £$; then Exit; elsego to step 2.

Where.

1. *u denotes the membership function*
2. *x demonstrates centroids*
3. *x illustrates the pixel intensity*
4. *£ is the termination criteria*

In this proposed system is implemented using this Fuzzy C-Means (FCM) clustering is a soft clustering technique where each data point belongs to multiple clusters with varying degrees of membership, represented by membership values [30]. The algorithm iteratively updates cluster centroids and calculates Euclidean distances to determine the degree of membership, minimizing the objective function based on these distances. Each centroid is computed as the weighted average of the data

points, with the weights determined by the membership values it is illustrated in Fig. 5. Through this, WBCs separates from other blood cells for the detection of lymphocytes.

3.5. Feature extraction

It is a significant point in the process of image classification. This helps in isolating the common properties of the image and converting the image into data so that it can process the data with the standard values which helps in identifying the infected image. Apart from the process of individual segmentation features required to train the models. The color features are extracted by converting the image to HSV then getting the values of HSV and then mean values of HSV can be calculated [31]. The Mean of the segmented image will be used as a threshold of the elements of a matrix.

$$Mean = mean2(Image) \tag{7}$$

$$Standard_Deviation = std2(Image) \tag{8}$$

The entropy is the state of concentration point that entity pixels can adapt which is used for qualitative analysis, evaluation, and comparison of image details.

$$Entropy = entropy(Image) \tag{9}$$

For finding the similarities between the two images, if there is no difference between any two images then the result of the (5) above function is zero.

$$RMS = mean2(rms(Image)) \tag{10}$$

The mean value gives the individual pixel intensity of the whole image. Variance is used for identifying how each pixel varies from the neighboring pixel.

$$Variance = mean2(var(double(Image))) \tag{11}$$

Contrast is the dissimilarity in the luminance that creates an object discernible from the other within the same ROI.

$$Contrast = \sum_{i,j=0}^{N-1} (i,j)^2 \, C(i,j) \tag{12}$$

$$Energy = \sum_{k=0}^{n} \binom{n}{k} x^k a^{n-k} \tag{13}$$

3.6. Image classification

The final stage is to compare the features of the input images with the features of the infected image which is done in this image classification stage. For performing this operation, the classifier is trained with features extracted from the previous stage. In this proposed system the classification is tested with two different Machine Learning Algorithms which are Support Vector Machine and Hybrid FCM and Random Forest Classification and the results are compared with other Machine learning algorithms also to find an efficient classifier for detecting the Acute Lymphocytic Leukemia which is demonstrated in Fig. 6.

SVM Support Vector Machine is the supervised Machine Learning Algorithm used for both the organization and regression of images. I perform well when have to perform binary classification. This falls under the machine learning category called kernel method and it is also called kernel machines. For performing this action feature extraction is done using Convolution Neural Networks (CNN) VGG 16 feature extraction. VGG 16 is a CNN that has sixteen layers deep, which is done by loading the pre-trained version of the trained network with millions of images through the ImageNet database. The Convolutional layers have 3 layers with 512 filters while the Max pooling layers to produce the SoftMax output from the input image which involves the fully connected layers of 4096 neurons. The feature extraction process has the pre-trained model by removing the fully connected layers, utilizing the convolutional layers as a feature extractor while the features are extracted from the final convolutional layer. This network can classify the images into various categories such as cars, animals, humans, mouse, keyboards, etc. as the network has learned the features of various images [32]. Fig. 7 represents the Support Vector Machine Classification; the main task is to find the hyperplane or the ideal line that separates the two classes.

Random Forest is a powerful tool used mainly for solving classification problems [33]. The most important disadvantage of the decision tree algorithm is the overfitting problem when the tree is deep the Random Forest classifier can limit the overfitting problem if there are more trees. The Random Forest is made of many trees more trees mean more strong forest. The Random Forest classifier gets the images from the dataset and creates a decision tree. The system gets the prediction from each decision tree and finally selects the best prediction based on the majority voting. The working process of the Random process algorithm is by selecting L random data points from the training data set. Then it constructs a tree corresponding to the selected L data points which is the subset. Choose the number of trees that needs to be constructed. These steps are iterated, and the results of each tree are evaluated and compared by majority voting in Fig. 8. The random forest is the combination of multiple numbers of trees to predict the class of the

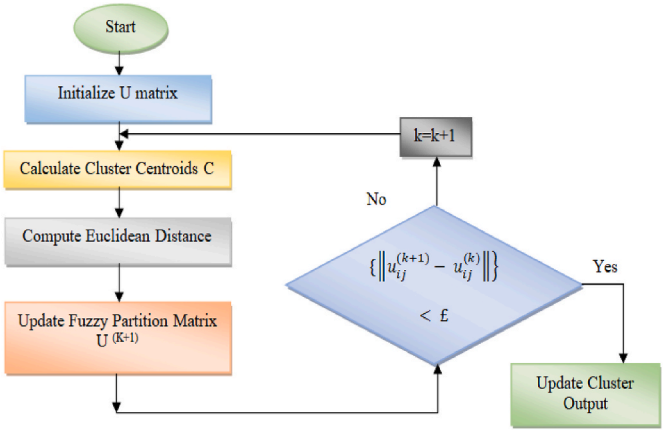


Fig. 5. Flowchart of FCM (fuzzy C means) clustering.

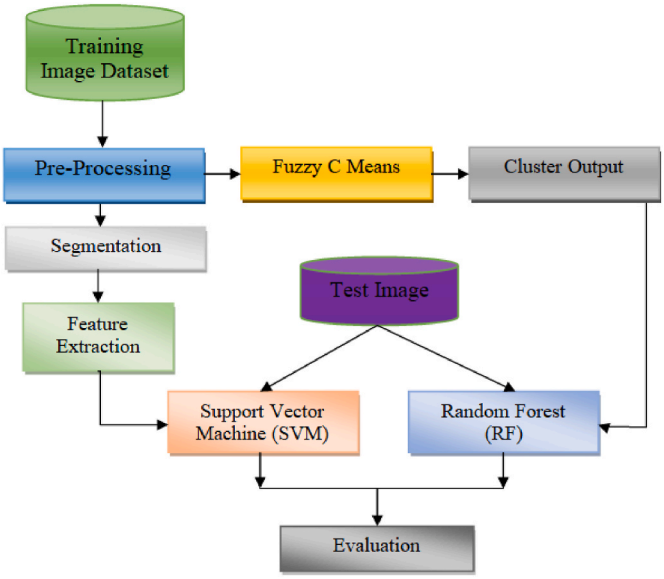


Fig. 6. Flowchart for the proposed system.

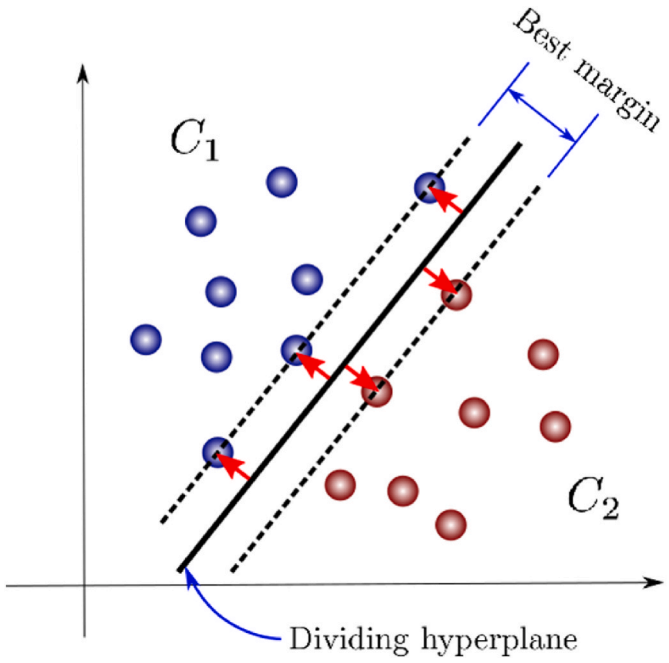


Fig. 7. Support vector machine classifier.

dataset. Some of the trees may give correct output and some may not, but when all the trees combine, they predict the class correctly. To make the random forest algorithm more efficient there should be the exact values in the dataset and the prediction results of each tree should have low correlations so that the random forest classifier can produce more accurate results than randomly predicted results this is achieved by Fuzzy C Mean Clustering algorithm which is used for extracting the actual values of the feature variable. The random forest classification algorithm is chosen for this proposed system because it takes very less time another major benefit of using a random forest classifier is it can maintain its accuracy when a large amount of data is unavailable or missing.

SVM was selected for its ability to handle high-dimensional data and find optimal classification boundaries, which is crucial for distinguishing between overlapping classes in leukemia detection. RF was chosen for its robustness in handling complex datasets and its ensemble approach, which improves classification accuracy by combining multiple decision trees. Together, RF generates a reliable feature set, and SVM enhances boundary precision, making the hybrid method highly effective for medical image classification. This combination provides better accuracy and stability than using either algorithm alone.

Algorithm Random Forest Classifier.

Step 1: Select Random L features from the features extracted from the image dataset.
Step 2: For Each y in L
IThe Information gain is calculated as
aGain (a, y) = E(a) – E(a,y) (14)
b $E(a) = \sum_{i=c}^{c} - P_i \, log_2 P_i$ (15)
c $E(a,y) = \sum_{c \in y} P(c)E(c)$ (16)
II Here E(a) is the entropy of two classes, and E(a,y) is the entropy of feature y.
III Select Node h which has the maximum information gain
IV Divide the nodes into sub-nodes
V Iterate steps I, II, and III to build the tree until it achieves the least number of samples necessary to be divided.
Step 3: Reiterate steps 1 and 2 N time for building the forest with N trees.

For selecting and optimizing the Random Forest algorithm, some parameters need to be tuned as the total number of decision trees, maximum depth of the decision tree, the minimum samples required to split the internal node, the features are required in every split to perform the bootstrap through the entropy value for splitting the criterion value while the weights are assigned to every class.

4. Results and discussion

The proposed method of classification of Acute Lymphocytic

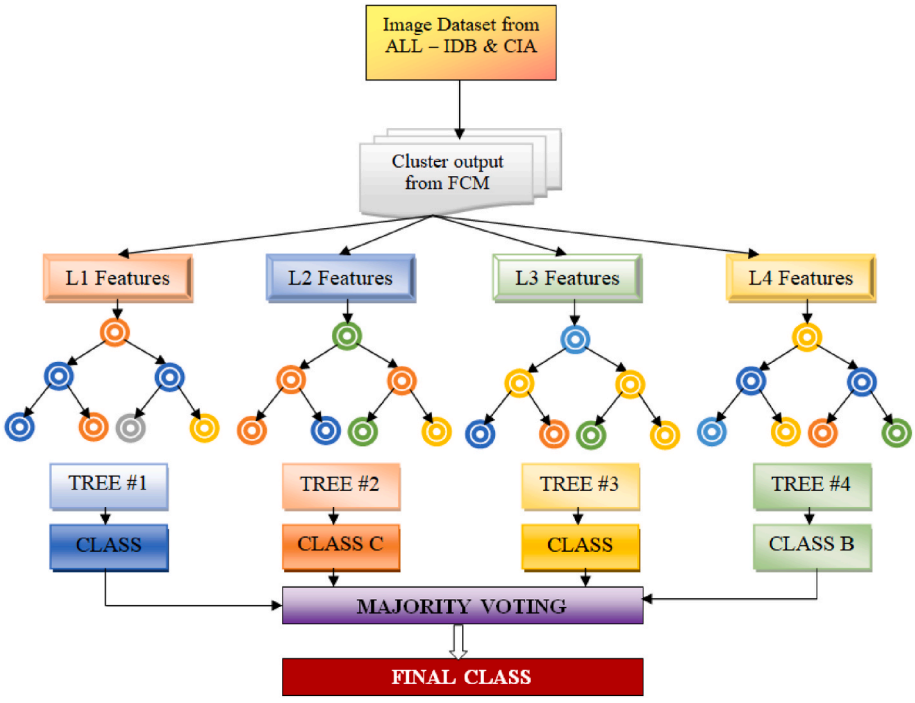


Fig. 8. Flowchart of random forest classifier.

Leukemia was performed in Matlab R2018 tools. The proposed methodology is compared with the related methodologies of VCGNet [7], GLCM [10], CNN [12], MBAML [25], and HALM [34]. The images from various sources such as ALL IDB and cancer imaging archives are initially augmented to improve the classification accuracy and to avoid overfitting problems. The flow of operation is given below.

Step 1: The image is pre-processed for removing the unwanted noise in the image, exposure, and lighting effects while extracting the blood samples through the microscope. This process is done by converting the image from SRGB to CMYK and histogram equalization.
Step 2: The image is segmented using fuzzy c means clustering through this image is divided into various clusters.
Step 3: Image features like colour and texture feature are extracted from the segmented images to perform the complex training operation.
Step 4: Using the SVM and Hybrid FCM RF classifier and finally the image is classified.
Step 5: The performance of two classifiers is compared with their ROC (Receiver Operating Characteristics).

Fig. 9 illustrates the image selected from the data set collected from various dataset libraries available online for performing the testing operation. Fig. 10 shows that the microscopic image selected from the dataset is loaded into the software for performing the classification operation and Fig. 11 illustrates the Image translation from RGB to CMYK format.

As the image in the dataset will be in the SRGB format it needs to be converted to CMYK format the purpose of this conversion is to differentiate WBC cells. As the image is a microscopic image there is a possibility of the image getting affected by needless noise and blurring. This could affect the quality of the image and at the same time, there is a possibility of a wrong diagnosis. For this purpose, the CMYK image is histogram equalized which is a very common technique to enhance the contrast of the image. This also overcomes all the lighting effects created during the image capture and Fig. 12 illustrates the Blood Microscopic Image Classified as Leukemia Detected.

Performance evaluation can be done through the confusion matrix in

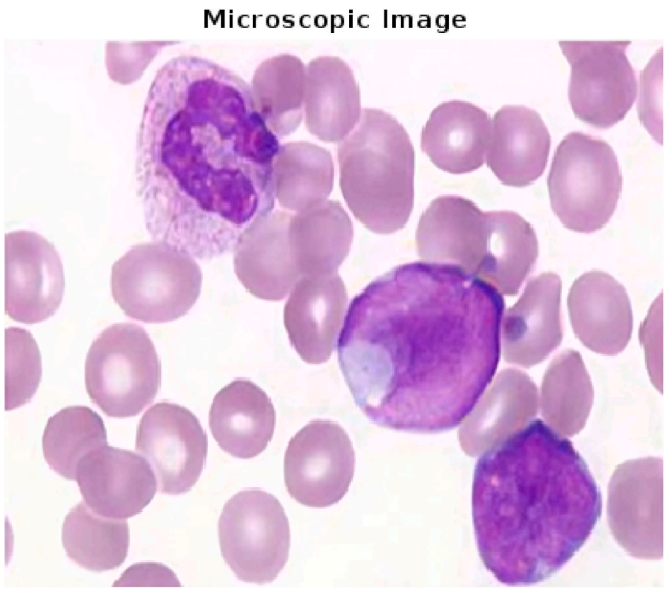


Fig. 10. Microscopic image loaded from the dataset.

Table 2, which is comprised of the four possible outcomes the True Positive (TP) which expects true, the real classification is true, True Negative (TN) which expects False; and the real classification is False, False Positive (FP), which expects True; But the real classification is False, and False Negative (FN) which expects False; and the real classification is True (see Table 3). The parameter comparion has proved that the proposed technique is performed well (see Table 4).

The performance metrics like specificity, sensitivity, Accuracy and are calculated using the formula.

$$Accuracy = \frac{TN + TP}{TN + FN + TP + FP} \times 100 \tag{17}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{18}$$

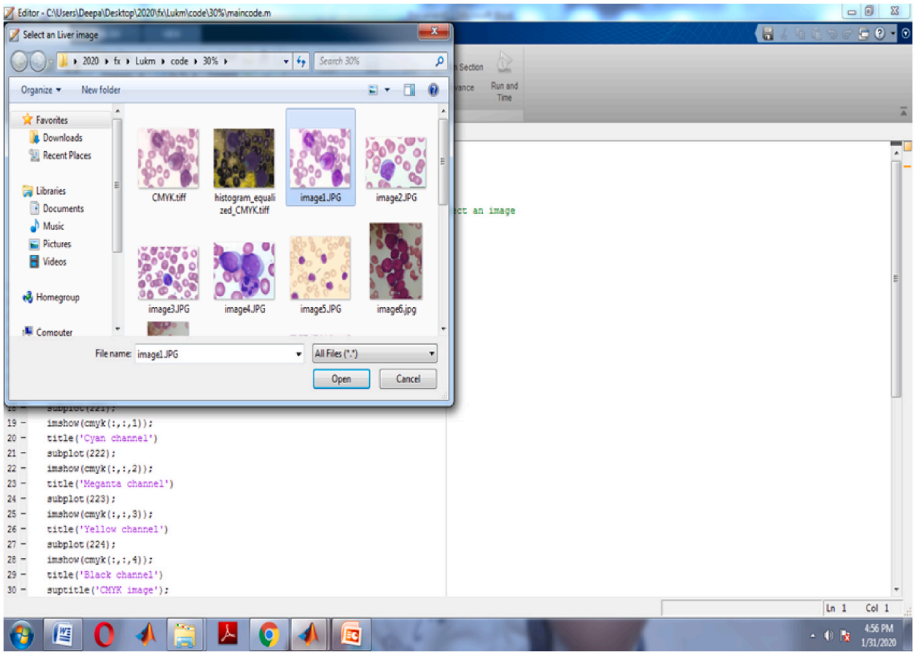


Fig. 9. Image Selection from the dataset.

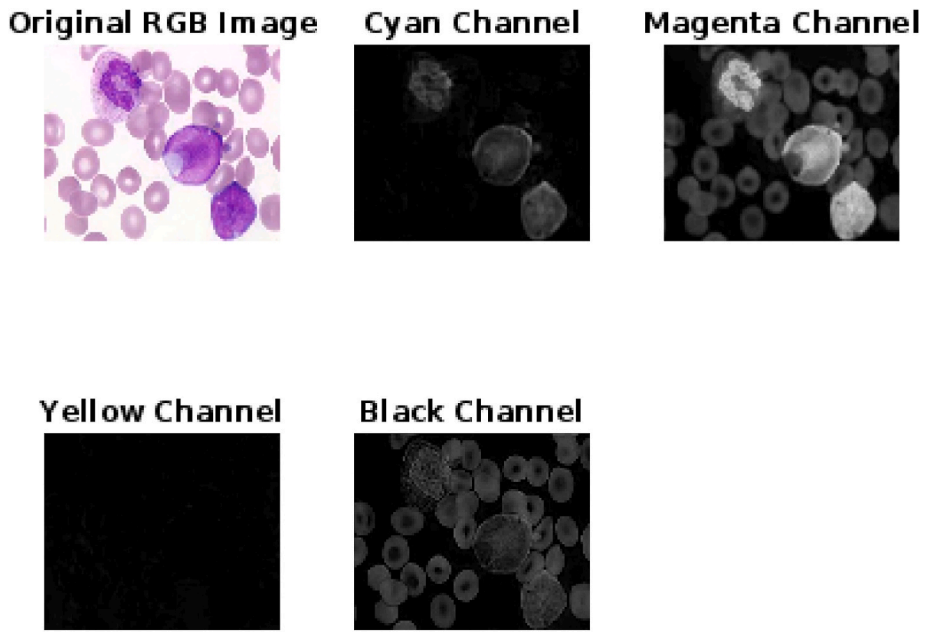


Fig. 11. Image translation from RGB to CMYK format.

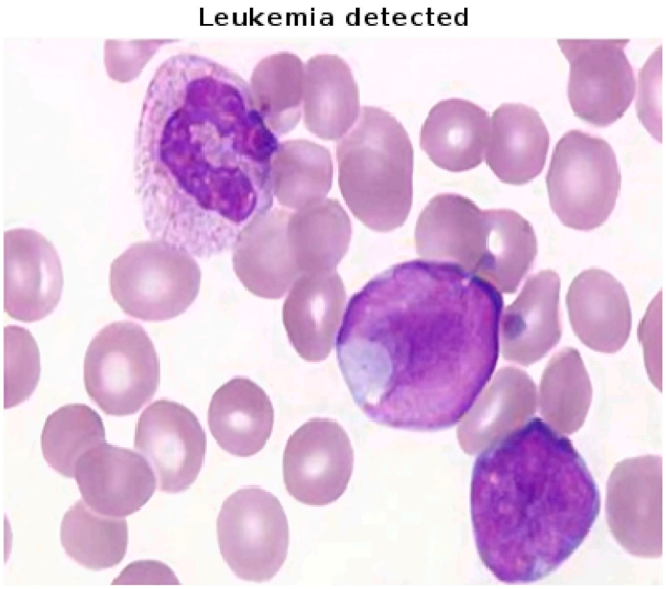


Fig. 12. Blood microscopic image classified as normal.

Table 2
Ratio of dataset.

S.NO	*Dataset*	*Training*	*Validation*	*Testing*
ALL IDB	*3885*	*2719*	*582*	*582*
Cancer Imaging *Archive*	*2075*	*1452*	*311*	*311*
Local Diagnostic *Centre*	*2677*	*1874*	*402*	*402*

$$Specificity = TN/(TN + FP) \tag{19}$$

Fig. 13 shows the ROC curve of the Random Forest classifier which is a graphical representation that shows the diagnostic capability of a classifier. ROC curve is created by plotting the curve of the True Positive against the False Positive at various thresholding settings. Similarly, the

Table 3
Confusion matrix.

Actual Classification	Predicated: Normal	Predicted: Leukemia Infected
Normal	True Negative (TN) 8600.29	False Positive (FP) 21
Leukemia Infected	False Negative (FN) 61	True Positive (TP) 8562

Table 4
Parameter comparison table.

S.NO	Algorithm	Sensitivity	Specificity	Accuracy
1	VGGNet	99.2	95	97
2	GLCM	98.3	98	98.2
3	CNN	98.4	97.4	97.1
4	MBAML	98.6	97.7	98.3
5	HALM	98.5	94.2	96.1
6	Proposed	99.4	97.8	99.06

ROC curve is plotted for the Support vector machine classifier too. The curve is close to the top left corner of the test is more efficient; with this on comparing both the plots itconclude that the Random Forest classifier is more effective for detecting Acute Lymphocytic Leukemia as the area under the curve of Random Forest (ROC).

Similarly, the datasets were compared with the Accuracy, Sensitivity, and Specificity of other machine learning algorithms like K Nearest Neighborhood (KNN), Decision tree, Convolution Neural networks (CNN), and Naive Bayes Algorithms experimental results demonstrate the superior performance of Random Forest classifier with Fuzzy C Means segmentation. Fig. 14 illustrates the comparison of Random Forest with other machine learning algorithms for ALL classification and the performance results proved that the proposed technique has enhanced result in terms of Sensitivity, Specificity, and Accuracy. Fig. 15 Shows the accuracy levels achieved under different training-to-testing dataset ratios. By varying these ratios, the figure provides insights into how the proportion of training data impacts the model's overall accuracy. A higher training-to-testing ratio generally leads to improved accuracy, as the model has more data to learn from, while a lower ratio may result in reduced accuracy due to insufficient training instances. Fig. 16 depicts the accuracy trends for both the training and

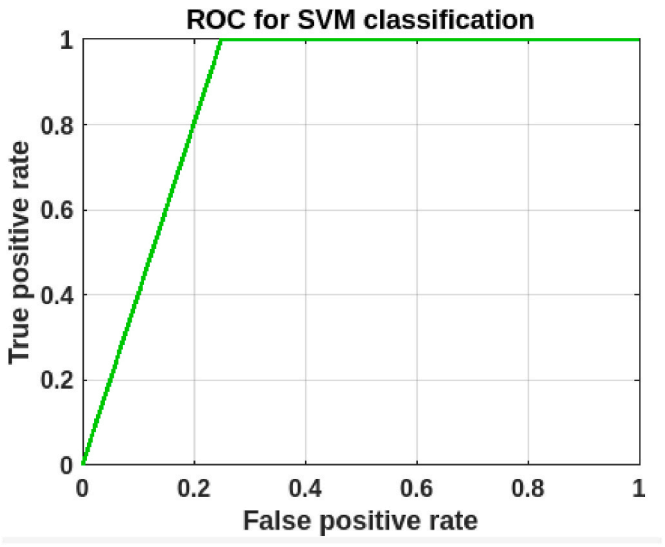


Fig. 13. Roc of support vector machine classifier.

validation datasets over the course of model training. This figure helps in evaluating the learning progress of the model, identifying potential overfitting or underfitting issues, model performs well on the training data but moderately on unseen data.

Fuzzy C-Means (FCM), Random Forest (RF), Support Vector Machine (SVM), and the hybrid RF + SVM offer different strengths in classification. FCM is a clustering algorithm that handles overlapping and ambiguous data well by assigning membership values, making it useful for medical images like leukemia detection. However, FCM is not a classifier and is sensitive to noise and initialization. RF, an ensemble classifier, combines multiple decision trees to provide strong classification performance, especially in complex datasets, but it can be computationally intensive. SVM is highly effective for high-dimensional data and finds an optimal hyperplane for classification, but it struggles with overlapping classes and is sensitive to outliers. The hybrid RF + SVM approach combines RF's ability to handle large, complex datasets with SVM's strength in optimizing classification boundaries. By using RF to generate a robust feature set and SVM to refine classification, the hybrid method offers superior accuracy and stability compared to standalone FCM, RF, or SVM. This makes RF + SVM particularly effective in medical image analysis, where overlapping data and the need for precise boundaries, such as in leukemia detection, require both robust feature handling and precise classification.

Whenever CNNs have enhanced performance on image classification process, it needs significant computational resources, the proposed work has maintained the domain-specific task which provides enhanced processing and low memory utilization. CNNs are parallelizing with GPU/TPU could accelerate training and batch interference.

5. Conclusions

Acute Lymphocytic Leukemia (ALL) is a critical form of cancer that necessitates early detection to prevent severe complications, including organ failure and increased mortality. This study presents a novel

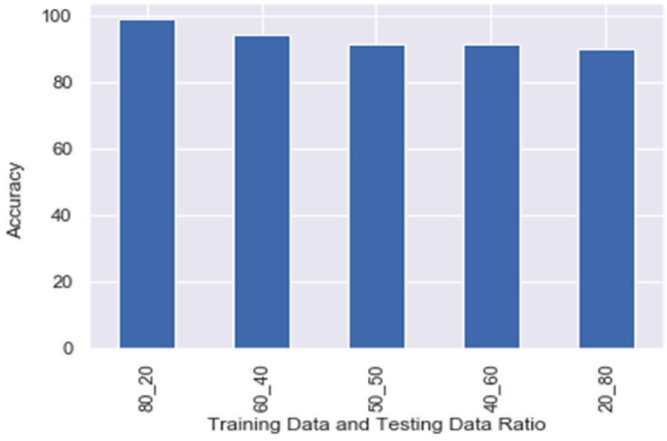


Fig. 15. Accuracy of different ratios of the Training and Testing dataset.

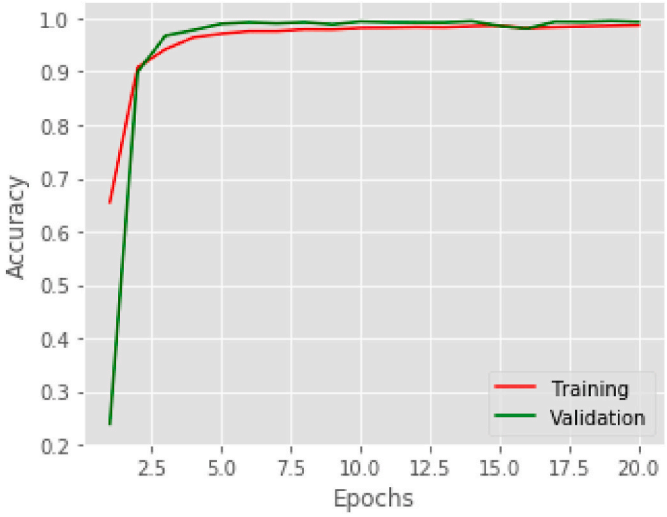


Fig. 16. Training and validation accuracy.

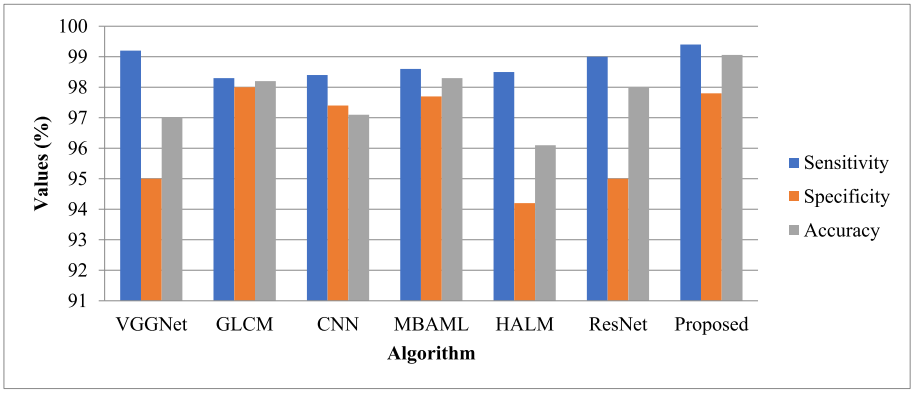


Fig. 14. Comparison of Random Forest with other machine learning algorithms for ALL classification.

approach utilizing a Hybrid Fuzzy C Means (FCM) clustering-based Random Forest (RF) Classifier for the effective classification of Acute Lymphocytic Leukemia into infected and normal categories. The proposed method demonstrated remarkable performance metrics, achieving an accuracy of 99.06 %, a sensitivity of 99.4 %, and a specificity of 97.8 %. These results indicate that the Hybrid FCM and RF Classifier not only surpass traditional manual diagnostic methods but also outperform several other machine learning algorithms, including Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Naive Bayes.

The high sensitivity and specificity rates imply that the proposed model can reliably identify both affected and non-affected individuals, minimizing the risk of false negatives and false positives. This capability is crucial in clinical settings, where accurate diagnostics can lead to timely intervention and improved patient outcomes. Moreover, the Receiver Operating Characteristic (ROC) curve analysis reinforces the effectiveness of the Hybrid FCM and RF Classifier as a robust tool for diagnosing Acute Lymphocytic Leukemia, providing clinicians with a reliable method for early detection. Looking ahead, future work will focus on the development of user-friendly applications for mobile and Windows platforms that enable the detection of various diseases through microscopic imaging of blood smears. Such advancements could democratize access to accurate diagnostic tools, especially in resource-limited settings, ultimately enhancing patient care and outcomes in the fight against leukemia and potentially other hematological disorders.

CRediT authorship contribution statement

K. Lakshmi Narayanan: Writing – original draft, Writing – review & editing, Conceptualization, Data curation. **R. Santhana Krishnan:** Writing – original draft, Conceptualization, Data curation. **Y. Harold Robinson:** Validation, Formal analysis, Supervision. **S. Vimal:** Conceptualization, Data curation. **Tarik A. Rashid:** Supervision, Data curation. **Chetna Kaushal:** Validation, Formal analysis. **Md. Mehedi Hassan:** Conceptualization, Supervision.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Dataset used

ImageNet dataset is used for the experiments and it is publicly available in https://paperswithcode.com/dataset/imagenet

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they do not have any conflict of interests. This research does not involve any human or animal participation. All authors have checked and agreed the submission.

Acknowledgment

The authors would like to express their appreciation to the Universities involved. The authors would like to thank the participants for their time in contributing to this study.

Data availability

Data will be made available on request.

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